PNP–BV Forward Solver Evolution
From 4-Species PNP to 3-Species + Boltzmann + Log-c + Log-Rate BV

Jake Weinstein

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1 Summary

The forward solver supporting the 4-parameter inverse problem $(\log k_{0,1}, \log k_{0,2}, \alpha_1, \alpha_2)$ was rebuilt over the past two weeks. Three independent changes, each motivated by a distinct numerical failure of the original 4-species PNP solver, are now stacked in the active configuration:

1. **4-species → 3-species PNP + analytic Boltzmann counterion** — removes the inert supporting-electrolyte counterion as a dynamic Nernst–Planck unknown while retaining its contribution to Poisson's equation.

2. **Concentration c_i → log-concentration $u_i = \ln c_i$** — enforces positivity of the dynamic species and avoids the negative concentration iterates that previously triggered floor/clip regularizations.

3. **Standard Butler–Volmer → log-rate Butler–Volmer** — removes a phantom R_2 sink at high anodic overpotential caused by clipping the surface concentration before evaluating the boundary reaction rate.

The combined effect is that the usable voltage grid expands from roughly $V_{\mathrm{RHE}} \leq 0.20\,\mathrm{V}$ to $V_{\mathrm{RHE}} \leq 0.60\,\mathrm{V}$. In the local Fisher analysis at TRUE parameter values, the old nearly singular $\log k_{0,2}$ direction is removed: $\mathrm{cond}(F)$ improves from about 2×10^{11} to about 1.79×10^{7}, and the canonical-ridge cosine drops from 1.000 to 0.031. This is a major improvement, but it should not be read as a complete solution of the inverse problem. Clean-data inverse runs still show multiple basins: α_1 and α_2 are recovered to $\leq 2\%$ in 2/4 standard initializations and to $\leq 16\%$ in 3/4, while no single initialization recovers all four parameters jointly.

2 Original 4-Species Solver

The original forward model was a coupled Poisson–Nernst–Planck (PNP) system on a graded 2D rectangular mesh with four species: O_2, H_2O_2, H^+, and the supporting-electrolyte counterion ClO_4^-. In dimensional form, the bulk equations for each species i are

$$\frac{\partial c_i}{\partial t} \;=\; -\nabla\cdot J_i, \qquad J_i = -D_i\left(\nabla c_i + \frac{z_i F}{RT} c_i \nabla\varphi\right), \qquad -\varepsilon\,\nabla^2\varphi \;=\; F\sum_i z_i c_i, \tag{1}$$

with Butler–Volmer (BV) flux boundary conditions at the electrode for the two reactions

$$\mathrm{R}_1:\ \mathrm{O_2} + 2\mathrm{H}^+ + 2e^- \rightarrow \mathrm{H_2O_2}, \qquad E_{\mathrm{eq}}^{(1)} = 0.68\,\mathrm{V}, \tag{2}$$

$$\mathrm{R}_2:\ \mathrm{H_2O_2} + 2\mathrm{H}^+ + 2e^- \rightarrow 2\mathrm{H_2O}, \qquad E_{\mathrm{eq}}^{(2)} = 1.78\,\mathrm{V}. \tag{3}$$

The implementation uses the corresponding nondimensional form, where voltage is scaled by $V_T = RT/F$.

Where it failed.

- **Voltage ceiling near $V_{\mathrm{RHE}} \approx 0.20\,\mathrm{V}$.** At higher anodic potentials the $\mathrm{H_2O_2}$ surface concentration can drop by many orders of magnitude. Newton iterates in the concentration formulation could become non-positive, triggering floor regularizations whose Jacobian contributions stalled the SNES solve.
- **Nearly singular FIM at TRUE.** Restricted to $V_{\mathrm{RHE}} \leq 0.20\,\mathrm{V}$, the CD+PC Fisher information had condition number $\approx 2 \times 10^{11}$ with weak eigenvector almost pure $\log k_{0,2}$. In that voltage window, $k_{0,2}$ was practically unidentifiable.
- **$\mathrm{ClO_4^-}$ added cost without chemistry.** The counterion is inert in this model and exists to provide charge balance. Solving for it as a fourth dynamic NP variable increased assembly and nonlinear-solve cost while not adding an electrode reaction or an independent inverse observable.

3 Change 1: 3-Species PNP + Analytic Boltzmann Counterion

Motivation. $\mathrm{ClO_4^-}$ is inert and present as supporting electrolyte. Rather than solving a fourth dynamic Nernst–Planck equation for this nonreactive counterion, the reduced model assumes it remains in local electrochemical equilibrium with the electrostatic potential. Under that assumption its profile is represented analytically by a Boltzmann factor, $c_{\mathrm{ClO_4}}(x) = c_{\mathrm{bulk}} \exp\bigl(-z_{\mathrm{ClO_4}}\varphi(x)/V_T\bigr)$, with the sign following the code's potential convention. This is a modeling reduction for an inert supporting electrolyte, not a new physical mechanism.

Implementation. Drop $\mathrm{ClO_4^-}$ as a primary unknown. The Poisson equation keeps its analytic contribution to the charge density:

$$-\varepsilon\nabla^2\varphi \;=\; F \sum_{i\in\{\mathrm{O_2},\,\mathrm{H_2O_2},\,\mathrm{H}^+\}} z_i c_i \;-\, F\, c_{\mathrm{ClO_4,bulk}}\, \exp(-\varphi/V_T), \tag{4}$$

where the exponential sign should be interpreted consistently with the nondimensional potential convention used in the code. The three remaining species have charges $z = [0, 0, +1]$ and retain the full NP coupling. Operationally this is implemented through the Boltzmann contribution to the Poisson residual; see `add_boltzmann()` in the current study scripts.

Result. The reduced model removes one dynamic species and produced a measurable speedup in the forward studies. To quantify how closely the reduced solver tracks the full 4-species PNP, the current production 3-species + Boltzmann log-c forward solver (the build path used by `v18_logc_lsq_inverse.py`, the most recent inverse) was run head-to-head against the 4-species standard solver via `solve_grid_with_charge_continuation` on a shared mesh, shared TRUE kinetic parameters, and shared E_{eq}. The 4-species charge-continuation pipeline reaches $z = 1$ across the documented window $V_{\mathrm{RHE}} \in [-0.50, +0.10]\,\mathrm{V}$, but the 3-species log-c solver only cold-starts in

$V_{\mathrm{RHE}} \gtrsim -0.20\,\mathrm{V}$. To grow the overlap set we add a voltage-continuation strategy on the 3-species side: cold-start where possible, then warm-start at full charge from the nearest converged neighbour and march the applied potential outward in ΔV substeps with bisection fallback. With four substeps per 0.10 V jump, this fills the entire $V_{\mathrm{RHE}} \in [-0.50, +0.10]\,\mathrm{V}$ grid. Across the resulting eight-voltage overlap set, both observables agree well within the F2 diffusion-limit tolerance (5% of $\max|\mathrm{obs}|$) that the inverse-study validation framework uses elsewhere (`Forward/bv_solver/validation.py`):

	max	**mean**
$\lvert\Delta\mathrm{CD}\rvert/\max\lvert\mathrm{CD}\rvert$	0.269%	0.201%
$\lvert\Delta\mathrm{PC}\rvert/\max\lvert\mathrm{PC}\rvert$	0.104%	0.033%

All eight overlap points pass the 5% F2 threshold on both observables. At anodic voltages where peroxide current is small ($V_{\mathrm{RHE}} \geq -0.10\,\mathrm{V}$), PC differences appear as a sign flip on values four to seven orders of magnitude below $\max|\mathrm{PC}| \approx 1.8 \times 10^{-1}$ — a numerical artefact of evaluating a near-zero quantity in two different formulations, not a physics disagreement. The important claim is therefore not that the two formulations are algebraically identical, but that this controlled reduction preserves the ORR observables to better than half a percent of the diffusion-limit scale across the entire window where the 4-species solver itself converges — well inside the accuracy needed for the current inverse-study comparisons. Raw data, CSV, plot, and summary are saved under `StudyResults/v24_3sp_logc_vs_4sp_validation/`; the comparison script (including the warm-start continuation strategy) is `scripts/studies/v24_3sp_logc_vs_4sp_validation.py`.

4 Change 2: Log-Concentration Formulation

Motivation. Concentrations near a reactive electrode can span many orders of magnitude. In the diffuse layer, $c_i(x)$ varies approximately as $c_{\mathrm{bulk}} \exp(-z_i\varphi/V_T)$. When c is the primary unknown, the SNES line search can overshoot near small concentrations and produce non-positive iterates. Those iterates then require floors or clips, and the regularization can contaminate the Jacobian and stall continuation.

Implementation. Switch to $u_i = \ln c_i$ as the primary unknown. The Nernst–Planck weak form becomes

$$\int_\Omega \frac{e^{u_i} - e^{u_i^{\mathrm{old}}}}{\Delta t}\, v \,\mathrm{d}x \;+\; \int_\Omega D_i\, e^{u_i}\big(\nabla u_i + z_i \nabla\phi\big)\cdot\nabla v \,\mathrm{d}x \;=\; (\text{BV boundary terms}), \tag{5}$$

and Poisson's right-hand side replaces c_i by $\exp(u_i)$. See `Forward/bv_solver/forms_logc.py`.

Why it helps.

- $c_i = \exp(u_i) > 0$ by construction, so the dynamic species remain positive in the bulk formulation.
- The Debye-layer behavior becomes approximately linear in u_i: $u_i(x) \approx \ln c_{\mathrm{bulk}} - z_i\varphi/V_T$, which is more naturally represented on the existing graded mesh.
- Remaining numerical guards/clamps should be understood as overflow protection rather than concentration floors in the physical variable; later clamp sweeps checked that these guards did not drive the reported Fisher results.

5 Change 3: Log-Rate Butler–Volmer

Starting point. The reference for our group's canonical BV expression is the standard two-branch form for an n-electron redox couple $\mathrm{O} + ne^- \rightleftharpoons \mathrm{R}$[1]:

$$R_{\mathrm{BV}} \;=\; k_0\Big[\, c_{\mathrm{O}}\, \exp\!\Big(-\alpha\, \tfrac{nF\eta}{RT}\Big) \;-\; c_{\mathrm{R}}\, \exp\!\Big((1-\alpha)\, \tfrac{nF\eta}{RT}\Big)\,\Big], \tag{6}$$

where $c_{\mathrm{O}}, c_{\mathrm{R}}$ are the surface concentrations on the electrode reaction plane and $\eta = \phi_m - \phi_s - E_{\mathrm{eq}}$ is the overpotential. In the nondimensional code, RT/F is replaced by V_T and $nF\eta/RT$ becomes $n\eta/V_T$. Equation (6) is the algebraic object both formulations evaluate; the two formulations disagree only on *how* the surface contribution is combined with the voltage exponential.

Two ways to evaluate it in a log-c solver. With $u_i = \ln c_i$ as the primary unknown, $c_{\mathrm{O}} = \exp(u_{\mathrm{O}})$ exactly. The two evaluations differ only at the boundary, in how u_i is turned back into a rate. Showing the cathodic branch only for clarity:

Old: clip-then-multiply	**New: log-rate**
$c_{\mathrm{surf},i} = \exp\big(\mathrm{clip}(u_i, \pm 30)\big)$	(no surface clip)
$r_j \;=\; k_{0,j}\, c_{\mathrm{surf,cat}(j)}\, \exp(-\alpha_j\, n_e\, \eta_j/V_T)$	$\log r_j \;=\; \ln k_{0,j} \;+\; u_{\mathrm{cat}(j)} \;-\; \alpha_j\, n_e\, \eta_j/V_T$
	$r_j \;=\; \exp(\log r_j)$
exponentiate u, clip, multiply	add u inside the exponent, exponentiate once

The anodic branch is analogous: replace $u_{\mathrm{cat}(j)}$ by $u_{\mathrm{anod}(j)}$ and the sign of the η term, then subtract the two exponentials. The full implementation also adds the $\sum_f \nu_f (u_{\mathrm{sp}(f)} - \ln c_{\mathrm{ref},f})$ contribution from cathodic concentration factors (e.g. the H^+ stoichiometric power in our $\mathrm{R}_1/\mathrm{R}_2$ reactions); under the new form those terms also enter additively inside the exponent.

Why the old form fails at high η. The two formulations agree algebraically when $|u_i|$ stays modest. They disagree when u_i runs to large negative values — exactly the regime this work was trying to access. In the old form, $u_{\mathrm{H_2O_2}}$ should tend to a very negative number at high anodic V_{RHE}, but the ± 30 clip pins $c_{\mathrm{surf,H_2O_2}}$ at a finite floor. Multiplying that floor by the large R_2 exponential produces a phantom cathodic sink. In the new form, the same large-negative u_i is added *inside* the exponent, so $\log r_j \to -\infty$ and $r_j \to 0$ smoothly, which is the physically correct behavior. This was a major source of Newton failures in the $V_{\mathrm{RHE}} \in [+0.30, +0.60]$ V range; the log-rate form removes them. The toggle is `bv_log_rate=True` in the BV convergence configuration; see `forms_logc.py:294--360`.

Result. The extended voltage grid $V_{\mathrm{RHE}} \in \{-0.10, +0.10, +0.20, +0.30, +0.40, +0.50, +0.60\}$ V now converges under the current cold-ramp/charge-continuation pipeline. Some points require the continuation machinery rather than a single direct Newton solve, so the safe statement is "converges under the current pipeline," not "converges trivially from every cold start."

[1] See the internal document `docs/PNP Equation Formulations.tex` for the group's full derivation, weak forms, and flux-boundary mapping. The form below is its R_{BV} expression.

6 Combined Results

Voltage window

Pre-rebuild, the usable synthetic voltage window was roughly $V_{\mathrm{RHE}} \leq 0.20\,\mathrm{V}$. Post-rebuild, the current pipeline reaches the extended grid through $V_{\mathrm{RHE}} = 0.60\,\mathrm{V}$. This matters because R_2 begins to unclip at approximately $V_{\mathrm{RHE}} > E_{\mathrm{eq}}^{(2)} - 50V_T \approx 0.495\,\mathrm{V}$, under the code's η_{scaled} clipping convention. The inverse problem now has access to voltage-shape information from a regime that was previously unreachable.

Fisher information

At TRUE parameter values, with current density (CD) and peroxide current (PC) as observables and a global-max 2% noise model:

	old grid ($\leq 0.20\,\mathrm{V}$)	**new grid** ($\leq 0.60\,\mathrm{V}$)
$\sigma_{\min}(S_{\mathrm{white}})$	2.35×10^{-2}	2.51
$\mathrm{cond}(F)$	2.0×10^{11}	1.79×10^{7}
canonical-ridge $\lvert\cos\rvert$	1.000	0.031
weak eigenvector	nearly pure $\log k_{0,2}$	almost pure $\log k_{0,1}$

The historic "$\log k_{0,2}$ unidentifiable" local Fisher result is gone. A new weak direction in $\log k_{0,1}$ remains. Thus the local FIM is much better conditioned than before, but it is still not perfectly conditioned, and it does not by itself guarantee global recovery from arbitrary initializations.

Adjoint validity

At three of the previously inaccessible voltages ($V_{\mathrm{RHE}} \in \{0.30, 0.40, 0.50\}$), an initial adjoint-vs-FD check appeared to fail by exactly a factor of two. Detailed sweeps over annotated SNES iteration count, steady-state tolerance, FD step size, and cold-ramp versus warm-start FD showed that the discrepancy came from the warm-start FD calculation, not from the adjoint. With cold-ramp finite differences, the relevant nonzero sensitivity components agree with the adjoint to about six significant figures.

Inverse-problem implications

With the new forward solver, the four-parameter trust-region inverse with clean data and four standard initializations produces:

- α_1, α_2 recovered to $\leq 2\%$ in 2/4 initializations and to $\leq 16\%$ in 3/4 initializations;
- each individual $k_{0,j}$ recovered to $\leq 5\%$ from at least one initialization;
- but no single initialization recovers all four parameters jointly.

The (k_0, α) Tafel-ridge structure is no longer a single-direction local singularity dominated by $\log k_{0,2}$. The remaining problem is a multi-basin global cost surface: the data constrain some kinetic combinations strongly, but the optimizer can still land in different Tafel-like basins depending on initialization.

7 What This Does and Does Not Solve

Solved or substantially improved.

- Forward simulation now reaches the extended synthetic voltage grid through 0.60 V under the current continuation pipeline.
- The old local Fisher weak direction dominated by $\log k_{0,2}$ is removed once log-rate BV and the extended grid are used.
- α_1 and α_2 are substantially more robustly determined than in the earlier single-experiment formulation, although recovery is still initialization-dependent.

Not solved.

- $\mathrm{cond}(F) \approx 1.79 \times 10^7$ is a major improvement over the previous $\approx 2 \times 10^{11}$, but it is not numerically "perfectly conditioned."
- The global cost surface for the four-parameter joint inverse contains multiple Tafel-ridge basins; the optimizer's landing basin depends on initialization.
- Soft priors at literature-defensible strength ($\sigma_{\log k_0} = \ln 3$ or $\ln 10$) shift within-basin position but do not bridge the inter-basin barriers observed in the current clean-data studies.

For the current inverse project, the forward solver is no longer the obvious primary bottleneck. The next-step plan is therefore to close out the single-experiment route with restart diagnostics and anchored Tafel-coordinate optimization, then move to multi-experiment Fisher-information design, starting with bulk-O_2 variation.

8 Relation to Prior Work

None of the three changes is methodologically novel in isolation; the contribution here is the engineering combination and its quantitative effect on Fisher conditioning for the ORR inverse problem. The relevant precedents, to the extent the literature was surveyed:

Log-concentration / log-density formulations (Change 2). The substitution $u_i = \ln c_i$ has a long history in adjacent fields. In semiconductor drift–diffusion, the closely related *Slotboom* (or quasi-Fermi-level) variables have been standard since the late 1960s, and the Scharfetter–Gummel finite-volume discretization built around them remains a default in device simulation [1]. For the Poisson–Nernst–Planck system specifically, Metti, Xu, and Liu introduced what they explicitly named the "log-density formulation" as a means of constructing positivity-preserving and energetically stable schemes [4]. Recent extensions push this idea to high-order space–time finite element methods with arbitrary order positivity and unconditional energy stability [5]. Our usage applies this established idea inside a Firedrake-based ORR PNP–BV solver; we do not claim novelty in the substitution itself.

Reduced supporting-electrolyte models (Change 1). Replacing the inert supporting electrolyte by an analytic Boltzmann profile in the Poisson equation while keeping the reactive species as full Nernst–Planck unknowns is a known model reduction commonly called the *Poisson–Boltzmann–Nernst–Planck* (PBNP) hybrid model [3]. It is used extensively for ion-channel and biomolecular

simulations, where most ionic species are passive bath and only a few are chemically active. Our 3sp-plus-Boltzmann setup adapts that reduction to a two-step ORR system; the modeling assumption (inert, locally-equilibrated supporting electrolyte) is the same.

Log-form rate evaluation (Change 3). Carrying chemical or electrochemical rates in log space to avoid floating overflow or underflow is standard in stiff combustion and plasma chemistry codes, where Arrhenius rates routinely span tens of orders of magnitude. In the electrochemistry literature, the well-known logarithmic form of the Butler–Volmer relation is the *Tafel* approximation [2], which drops one of the two BV branches in the high-overpotential limit and is therefore mathematically distinct from the substitution used here. The present writeup retains both anodic and cathodic branches, but evaluates each branch as $r_j = \exp(\log r_j)$ inside the boundary residual so that the unclamped log-concentrations enter additively. We did not locate a specific cited precedent for that exact construction in the BV electrochemistry literature; it may be a small local numerical novelty, although the underlying log-form-rate idea is not.

Tafel approximation versus log-rate BV. To make the distinction precise: the Tafel form is a *physical approximation* valid only at large $|\eta|$, obtained by dropping the exponentially smaller BV branch; the log-rate BV evaluation is an *algebraic identity* that keeps both branches at every η, including the cross-over regime $\eta \to 0$ where Tafel is invalid. Both forms incidentally avoid the boundary-level surface-concentration clip at high $|\eta|$—Tafel because only one exponential remains, log-rate BV because the relevant u_i enters additively before exponentiation—but the two are not interchangeable. A forward PNP–BV solver that must run across a voltage grid containing $\eta \approx 0$ transitions needs the identity, not the asymptote. Conversely, when fitting a Tafel slope from experimental log-current data, the Tafel approximation is the appropriate construction; log-rate BV would be the wrong layer of the modeling stack.

Code Pointers

- Log-c PNP weak forms: `Forward/bv_solver/forms_logc.py`. Key functions include `build_context_logc`, `build_forms_logc`, and `set_initial_conditions_logc`.
- Log-rate BV toggle: `bv_log_rate` in `Forward/bv_solver/config.py`; consumed in `forms_logc.py`, around lines 294–360.
- Boltzmann counterion residual: `add_boltzmann()` in the top-level study scripts, including `scripts/studies/v18_logc_lsq_inverse.py`.
- Charge-continuation pipeline: `Forward/bv_solver/grid_charge_continuation.py`.

A Per-voltage 4sp vs 3sp+Boltzmann log-c overlap data

Per-voltage observable values from `StudyResults/v24_3sp_logc_vs_4sp_validation/comparison.csv`, generated by `scripts/studies/v24_3sp_logc_vs_4sp_validation.py`. Both solvers share the same graded mesh (Nx= 8, Ny= 200, $\beta = 3$), the same TRUE kinetic parameters ($\hat{k}_{0,1} = 1.263158 \times 10^{-3}$, $\hat{k}_{0,2} = 5.263158 \times 10^{-5}$, $\alpha_1 = 0.627$, $\alpha_2 = 0.5$), the same equilibrium potentials ($E_{\mathrm{eq}}^{(1)} = 0.68\,\mathrm{V}$, $E_{\mathrm{eq}}^{(2)} = 1.78\,\mathrm{V}$ vs RHE), and the same I_{SCALE}. The 4-species side runs through `solve_grid_with_charge_continuation`; the 3-species log-c side runs the production build path from `v18_logc_lsq_inverse.py`, cold-starting where possible and warm-starting (with paf substepping plus bisection) from the nearest converged neighbour everywhere else. Errors are reported

relative to max |obs| on the eight-voltage grid: $\max|\mathrm{CD}| \approx 1.84 \times 10^{-1}$, $\max|\mathrm{PC}| \approx 1.81 \times 10^{-1}$ (in nondimensional flux units; both solvers convert to mA/cm^2 via $-I_{\mathrm{SCALE}}$).

V_{RHE} (V)	$\mathrm{CD}_{4\mathrm{sp}}$	$\mathrm{CD}_{3\mathrm{sp}}$	$\lvert\Delta\mathrm{CD}\rvert/\mathrm{CD}_{\max}$ %	$\mathrm{PC}_{4\mathrm{sp}}$	$\mathrm{PC}_{3\mathrm{sp}}$	$\lvert\Delta\mathrm{PC}\rvert/\mathrm{PC}_{\max}$ %	3sp method
−0.500	−1.8390e−1	−1.8410e−1	0.108	−1.8112e−1	−1.8131e−1	0.104	warm← −0.40
−0.400	−1.8283e−1	−1.8305e−1	0.122	−1.6374e−1	−1.6391e−1	0.094	warm← −0.30
−0.300	−1.8133e−1	−1.8172e−1	0.212	−6.4999e−2	−6.5039e−2	0.022	warm← −0.20
−0.200	−1.7968e−1	−1.8017e−1	0.268	−1.5838e−3	−1.5678e−3	0.009	cold
−0.100	−1.7753e−1	−1.7802e−1	0.266	−1.2483e−5	+2.9058e−6	0.008	cold
+0.000	−1.7330e−1	−1.7379e−1	0.266	−9.6937e−8	+1.5288e−5	0.008	cold
+0.050	−1.6936e−1	−1.6985e−1	0.269	−8.6209e−9	+1.5378e−5	0.008	cold
+0.100	−1.6289e−1	−1.6307e−1	0.100	−7.7618e−10	+1.5388e−5	0.008	cold
		aggregate	**0.269** (max) 0.201 (mean)			**0.104** (max) 0.033 (mean)	8/

Reading the table. CD agreement stays under 0.27% of $\mathrm{CD}_{\max}$ across the full eight-voltage grid, including the three cathodic points reached only by warm-start continuation. PC agreement stays under 0.11% of $\mathrm{PC}_{\max}$ everywhere. The sign flips on PC at $V_{\mathrm{RHE}} \geq -0.10\,\mathrm{V}$ correspond to PC values four to seven orders of magnitude below $\mathrm{PC}_{\max}$ — a numerical artefact of evaluating a near-zero quantity in two different formulations, not a physics disagreement. Wall time on this run was 17.8 s for the 4-species sweep and 74.3 s for the 3-species cold/warm pipeline.

References

[1] Scharfetter, D. L., and Gummel, H. K., *Large-signal analysis of a silicon Read diode oscillator*, IEEE Trans. Electron Devices, **16**(1), 64–77 (1969). The original drift-diffusion finite-volume discretization built around Slotboom-type variables; widely cited as the canonical positivity-preserving scheme for charge-transport problems mathematically equivalent to PNP.

[2] Tafel, J., *Über die Polarisation bei kathodischer Wasserstoffentwicklung*, Z. Phys. Chem. **50**, 641–712 (1905). Original derivation of the high-overpotential logarithmic relation between overpotential and current that bears Tafel's name; mathematically distinct from the log-rate BV evaluation used here, but the historical antecedent for "log-form" relations in electrochemistry.

[3] Zheng, Q., Chen, D., and Wei, G. W., *Second-order Poisson–Nernst–Planck solver for ion transport*, J. Comput. Phys. **230**(13), 5239–5262 (2011); and the closely related "Poisson–Boltzmann–Nernst–Planck model," J. Chem. Phys. **134**, 194101 (2011), doi:10.1063/1.3581031. Establishes the PBNP hybrid: dynamic NP for chemically active species, Boltzmann factor for passive supporting electrolyte.

[4] Metti, M. S., Xu, J., and Liu, C., *Energetically stable discretizations for charge transport and electrokinetic models*, J. Comput. Phys. **306**, 1–18 (2016), doi:10.1016/j.jcp.2015.10.053. Introduces the substitution $u_i = \ln c_i$ for PNP under the explicit name *log-density formulation*, with positivity preservation and free-energy dissipation as the headline numerical properties.

[5] Liu, H., Maxian, O., Shen, J., Zitelli, J., Waltz, R. A., and Monteiro, P. B., *High-order space–time finite element methods for the Poisson–Nernst–Planck equations: Positivity and unconditional energy stability*, Comput. Methods Appl. Mech. Engrg. **396**, 115070 (2022); arXiv:2105.01163.

Extends the Metti et al. entropy/log-density formulation to arbitrary-order space–time FEM with positivity preservation and unconditional energy stability.